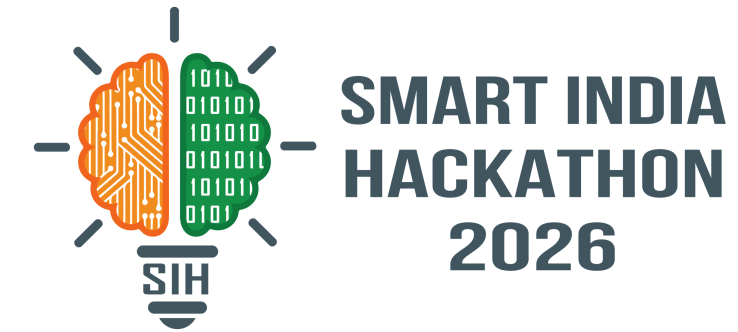


SMART INDIA HACKATHON 2026

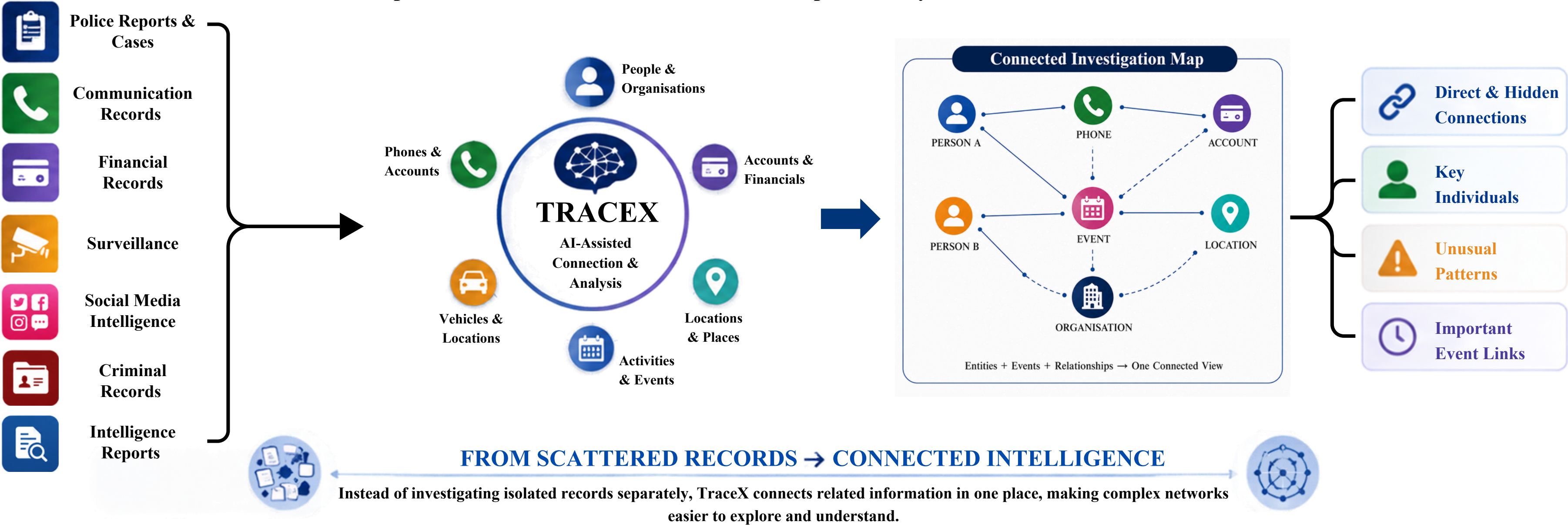


- **Problem Statement ID – 26189**
- **Problem Statement Title - AI Powered Criminal Network Analysis System**
- **Theme - Blockchain & Cybersecurity**
- **PS Category - Software**
- **Team ID - [Enter Team ID]**
- **Team Name - TraceX**



AI-Powered Criminal Network Analysis & Investigation Intelligence Platform

TraceX brings scattered investigation data together and uses AI-assisted analysis to connect important information and reveal relationships that may otherwise remain hidden.



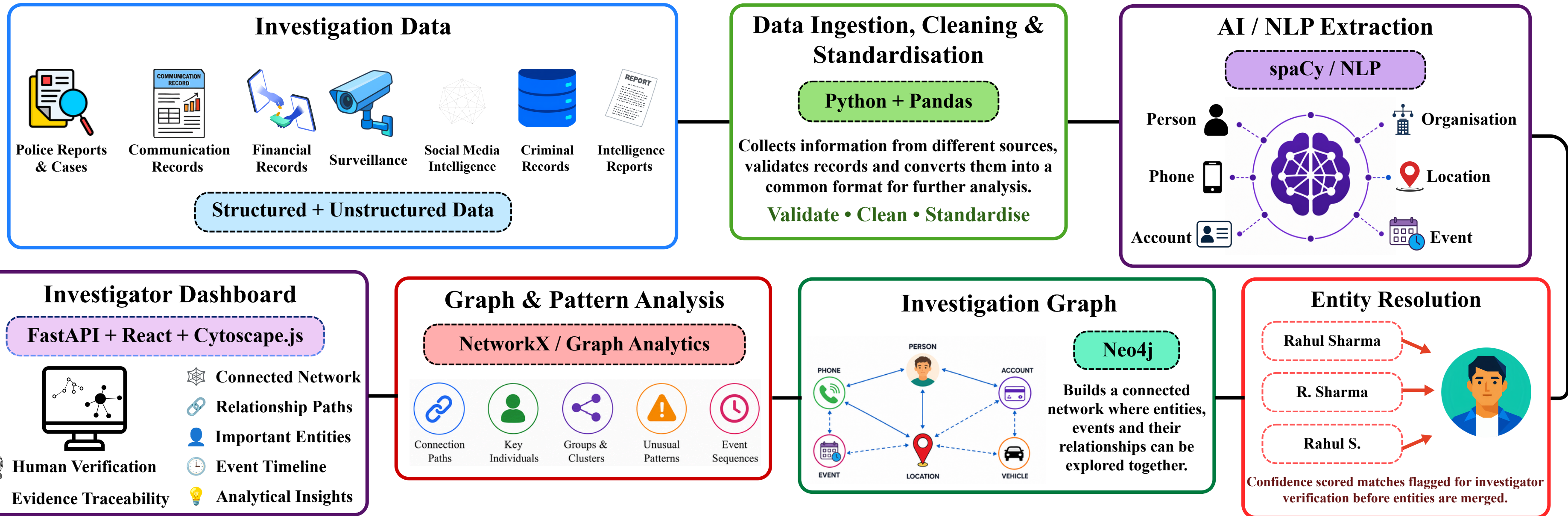
Multiple Data Types
Connects financial, communication, cyber and physical investigation information.

Relationship-First Approach
Focuses not only on individual records but also on how people, events and other entities are connected.

AI-Assisted & Insights
Helps surface important patterns while allowing Investigators to explore the relationships and events behind them.

Flexible Investigation Model
Designed to support different investigation scenarios instead of one specific crime type.

Secure & Auditable
Every connection is traceable to its source, with tamper-evident logs supporting evidence and courtroom admissibility.



Technology Stack



WHY TRACEX IS FEASIBLE

1 Validated Data Foundation

A large-scale investigation dataset was prepared from real-world data and enriched to address missing information.

50K+

Persons

1M+

Transactions

1.5M+

Comm. Events

750K+

Network Events

1M+

Relationships

2,000

Invest. Cases

Validation Status: PASS Structured and scenario validation completed successfully across all defined test cases.

3 Safe Data Foundation



Sensitive
Real Data



Not Required
for MVP



Realistic
Synthetic Data



Safe Development
& Testing

Prototype development without depending on sensitive criminal records.

Design aligned with DPDP Act 2023 principles for data minimisation and privacy-by-design.

2 Modular Architecture

TraceX can be developed and improved in independent stages:



Data
Processing



AI / NLP



Invest.
Graph



Analytics



Dashboard

Each layer can be built, tested and improved independently.

4 Scalable Approach

The modular architecture allows data processing, graph storage and analytics components to scale independently as investigation data grows.



Small Dataset



Larger Dataset



Connected
Ecosystem

KEY CHALLENGES & MITIGATION STRATEGIES



Fragmented
Data Formats



Common data model and data
standardisation pipeline



Duplicate or
Ambiguous Entities



Entity resolution and confidence-
based matching



Data Quality &
Reliability



Data validation, confidence scores and
source/provenance tracking



Large Data
Volume



Neo4j + NetworkX for prototype
analysis, scalable to Neo4j GDS for
larger datasets.



Unstructure
d Documents



NLP-assisted information extraction
with validation



Sensitive
Interpretation



Explainable insights with supporting
relationships and investigator review



Privacy &
Controlled
Access



Role based access, encryption, and blockchain
audit logs for secure, tamper-proof chain of
custody.

PROGRESSIVE DEVELOPMENT PATH

PHASE 1



FOUNDATION

Validated Data Model + Synthetic Investigation Dataset



PHASE 2



CORE PROTOTYPE

Data Ingestion + Data Processing + Entity Extraction +
Investigation Graph



PHASE 3



INTELLIGENCE LAYER

Graph Analytics + Pattern Detection + Explainable
Investigator Dashboard

IMPACT AND BENEFITS

From Fragmented Data to Better Investigation Support

TARGET AUDIENCE



IMPACT ON THE TARGET AUDIENCE



Faster Understanding of Complex Cases

Connected information helps investigators understand relationships across multiple records in one view.



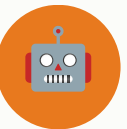
Discovery of Direct & Hidden Connections

Links between people, communication, transactions, locations and events become easier to identify and explore.



Faster Cross-Source Analysis

Information from different records can be connected instead of being analysed separately.



Reduced Manual Analysis

AI-assisted processing reduces repetitive effort involved in reviewing and connecting large volumes of information.



Better Prioritisation

Important entities, connections and unusual patterns help investigators focus on areas requiring further review.



Clearer Evidence Context

Insights can be explored along with their supporting relationships, records and events instead of appearing as isolated outputs.



OPERATIONAL & ORGANISATIONAL BENEFITS



Brings multiple investigation data sources into one connected view.



Provides a shared network view for easier review of complex information.



Keeps relationships and historical context easier to track over time.



Captures investigation connections and context within the system instead of relying only on memory.



BROADER BENEFITS



Strengthens NCRB's mission to protect women's safety by helping investigators uncover trafficking networks, repeat-offender patterns, and organised crime rings targeting women more quickly and reliably.



Protects sensitive investigation data through encrypted storage, controlled access, and tamper-evident audit trails aligned with data protection regulations including a blockchain-backed evidence trail for court-admissible chain of custody.



Saves time and resources by reducing repetitive manual work.



LONG-TERM POTENTIAL



Supports organised crime, financial crime, cybercrime and other network-based investigations.



The architecture can expand over time to handle larger datasets and integrate additional investigation data sources.



Highlights growing clusters, new relationships and unusual patterns for investigator review.



OVERALL IMPACT: **SCATTERED DATA** → **CONNECTED UNDERSTANDING** → **BETTER INVESTIGATION SUPPORT**





DATA SOURCES

1

National Crime Records Bureau (NCRB), India Crime in India 2023

https://www.data.gov.in/?utm_source

2

VAST Challenge 2009 — Visual Analytics Benchmark Dataset

https://vast.cs.umass.edu/VAST%20Challenge%202009/challenges/Grand%20Challenge%202009/?utm_source

3

ULB Credit Card Fraud Detection Dataset

<https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud/data>

4

Entity Extraction of Key Elements in Police Reports Based on Large Language Models (2024)

https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi%3A10.7910%2FDVN%2FEGZHFV&utm_source



RESEARCH REFERENCES

1

Berlusconi et al. — Link Prediction in Criminal Networks: A Tool for Criminal Intelligence Analysis

https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0154244&utm_source

2

Inductive and Transductive Link Prediction for Criminal Network Analysis (2023)

https://www.sciencedirect.com/science/article/pii/S1877750323001230?utm_source

3

Binette & Steorts — (Almost) All of Entity Resolution

https://pmc.ncbi.nlm.nih.gov/articles/PMC11636688/?utm_source

4

Entity Extraction of Key Elements in Police Reports Based on Large Language Models (2024)

https://www.mdpi.com/2076-3417/14/17/7819?utm_source